

DEPARTURE FROM THE PHOTON BASELINE: PAPER 11

PDF CHEAT SHEET

The core physics — Level A results, definitions, and figures.
Defensive armor removed: no sweep descriptions, no audit trail, no changelog.

Paper 11: Simple States and Complex Configurations
v69 (2026-10-07) — By: Dervis Kadunic

1 THE THESIS

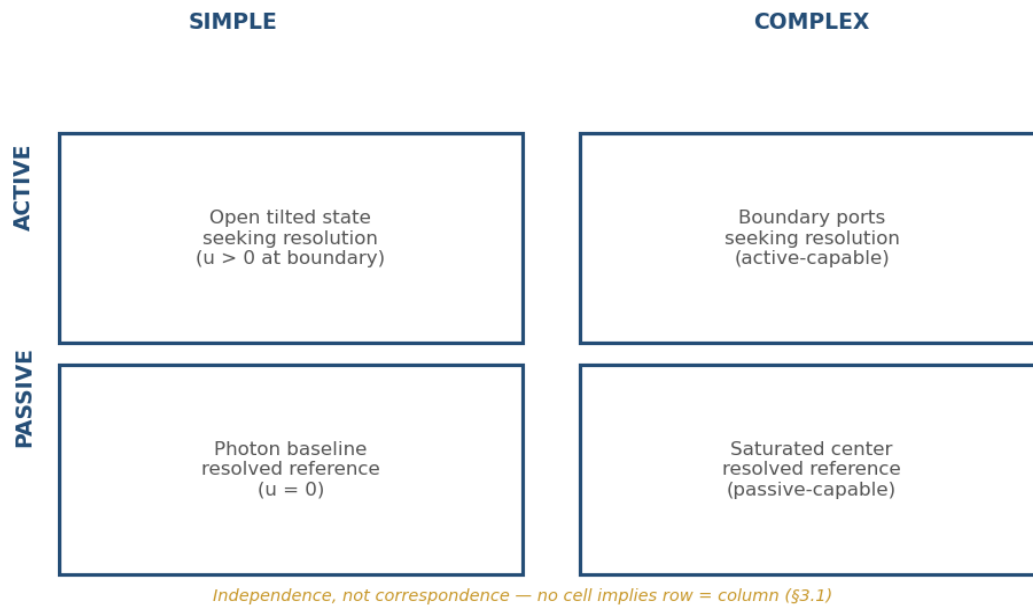
Paper 11 formalizes the role structure latent since Paper 2: what performs resolution, what serves as its reference, and at which level of organization each role appears. No new particles, no new ontology — only the relational structure already latent in the 27-state inventory, made explicit.

Two orthogonal cuts:

- Structural: a simple state is an individual configuration from the 27, a single node carrying its tilt triple; a complex configuration is a bonded aggregate of simple states — atomic configurations and above — carrying open ports of unresolved boundary geometry.
- Operational: the active role is played by an unresolved aspect performing resolution against the photon baseline; the passive role is played by a resolved aspect serving as the reference toward which resolution proceeds.

The cuts are orthogonal, not identical. A simple open state can play the active role; a complex interior can play the passive role. Collapsing "simple = passive" or "complex = active" into identities is the error this paper's formalization prevents.

FIGURE 1 — THE TWO CUTS



2 NATIVE ONTOLOGY

2.1 The primitives

Tick. The Stage-N native event identifier — the serialization of the scan ("this is from this"). Physical-scale mapping belongs to Stage P.

a_k . The tick index ordering ledger events — scan cycles and state changes alike. Forward-accumulating, strictly cumulative, non-invertible (Paper 2 §4.1, §12.1, §12.5).

Port. A boundary position where a configuration presents values to the scan. Each node has exactly six ports (three axes $\times$ two directions).

Junction. A meeting point where ports from configurations converge. Arity $k_j \geq 2$.

Aspect. Four kinds: port, junction, node, whole-configuration (§4.2). Roles attach to aspects, never to whole configurations intrinsically.

Unmatched integers (u). The count of presented values at an aspect not yet matched to a complementary value. Read per-tick off the presented junction values — no stored quantity.

2.2 The 27-state inventory

Paper 2's 27 configurations, each a tilt triple over three axes with values in {X, Y, O} (X-dominant, Y-dominant, neutral). State 9 is the photon baseline (O,O,O) — the universal ruler. States 1 and 9 have saturated centers: no unresolved integer at the center (Paper 2 §11.1).

The Per-Triangle Rule: exactly one X, one Y, one O per triangle (§0.5.w). The Circa budget: each aspect re-derives its fixed budget every tick (Paper 10 §2.2).

2b THE 27 STATES (CONDENSED)

Each state is a tilt triple over three axes. X = X-dominant, Y = Y-dominant, O = neutral/balanced. The photon (State 9: O,O,O) is the baseline.

- States 1–8: Single-axis tilts and their complements: (X,X,X), (Y,Y,Y), and mixed X/Y/O triples.
- State 9: Photon baseline (O,O,O) — the universal ruler; saturated center.
- States 10–18: Two-axis tilt combinations; varying X/Y dominance patterns.
- States 19–27: Three-axis complex tilts; the most departed from baseline.

Saturated centers (States 1 and 9): no unresolved integer at the center — "their geometric structure is settled" (Paper 2 §11.1). These can play the passive role as references.

Outside Content: a simple state's boundary tuple is its presented port values — the fixed measuring stick for encounters (Paper 3 §2).

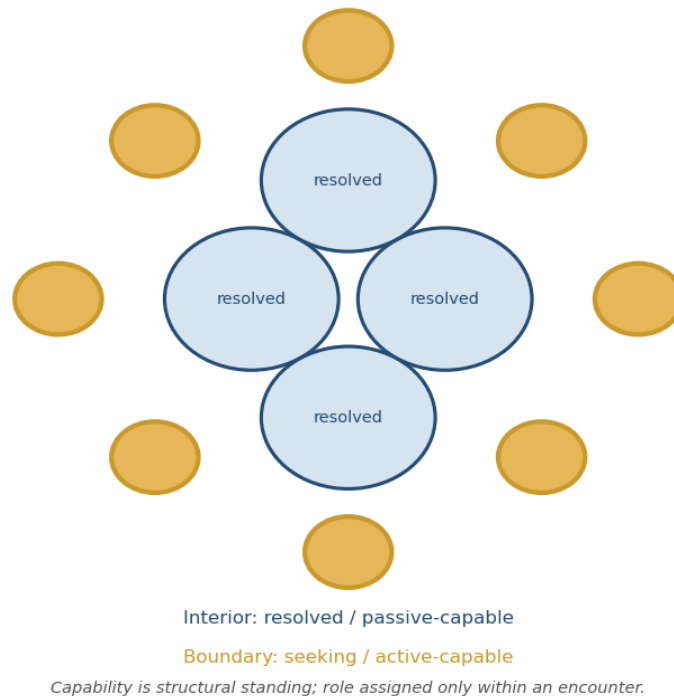
3 PORT STATUS AND PARTITION EXACTNESS

Definition (Port Status). At any tick, a port is open if the junction at that port is unresolved, and closed if the junction is resolved. Every port is exactly one of these two: a junction is either completed by the bonding modes, consuming open status (§6.1), or not. A port admits at most one completed junction.

Lemma (Port-Partition Exactness) [LEVEL A] [L6-P11] (§4.3). The set of consumed ports partitions exactly: completions at distinct a_k ledger entries are disjoint by the "no longer open" persistence (§6.1); completions within one entry are disjoint by the Port Status binary; no port appears twice in one junction's consumed set. Two scans cannot allocate the same finite contribution twice without manufacturing extra geometry.

Definition (Compound Closed-Loop). A complex configuration whose interior junction-aspects are resolved (passive-capable at the center) while its boundary ports remain open (active-capable at the boundary).

FIGURE 2 — THE COMPOUND CLOSED-LOOP



4 THE RELATIONAL ROLE THEOREM

Definition (Active/Passive). In an encounter R at tick τ : aspect a is active iff it presents unmatched integers ($u(a, \tau) > 0$) at an open port; aspect b is passive iff it serves as the resolved reference ($u = 0$ at the contact, toward whose presented values the shove matches).

Theorem (Relational Roles) [LEVEL A] (§3.1). No configuration is intrinsically active or passive. Activity and passivity are exact operational labels assigned to aspect-port states at a given tick. Proof. F.39 fixes one persistent selector and exhibits it active in R_1 ($u > 0$) and passive in R_2 ($u = 0$) — the selector literally identical across ticks while only port values change. Since the same aspect satisfies and fails the predicate, no configuration carries its role intrinsically. ■

Corollary (Capability-Actuality). "Active"/"passive" name roles, never standing states. Capability is marked by "can play" / "candidate for"; actuality requires encounter-indexing. Standing states are "seeking" ($u > 0$) or "resolved" ($u = 0$).

5 THE RESOLUTION VERB

Unmatched integers produce geometric shear tension on the fabric ($\tau_j = \Sigma x^2$ over the junction), which produces a mechanical shove into alignment, which produces relief toward the zero-tension baseline.

Definition (Alignment) [LEVEL A] [L1] (§7.2). A junction is aligned iff $\tau_j = 0$ (all integers matched; PTR-compatible). A transition is toward alignment iff it strictly decreases τ_j per tick. "Toward" names per-tick direction, not guaranteed arrival.

The transition operator T (§7.5) is scoped to pairwise junctions ($k_j = 2$): local state $\sigma = (A, B)$, each action zeroes the canonically-first complementary pair, $\Delta\tau_j = -2$ per tick. An arity-3 junction does not resolve via T .

Triple-Convergence Resolution [LEVEL A] [inherited] (§7.6). Paper 2 §8.1: when three State-4 configurations converge at their centers, they lock into a special bond; on completion the tension releases as three outputs — a

photon, an electron, and a positron. Nine X, nine Y, nine O in and out; nothing created or destroyed.

FIGURE 3 — THE RESOLUTION VERB (STATE TRANSITION)



Ledger states, not objects moving through space.

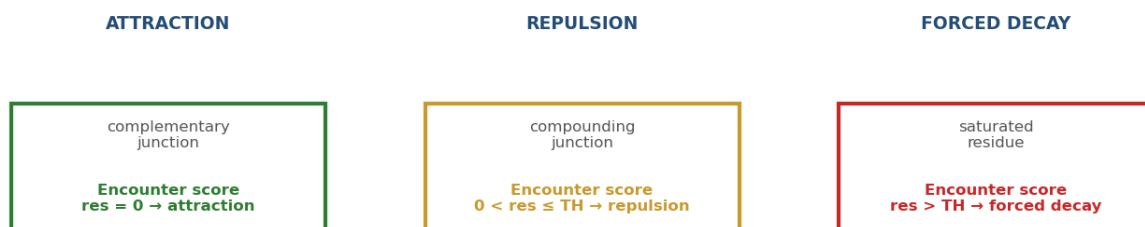
Stage-P observational shadow: spatial convergence

6 THE INTERACTION TAXONOMY

Encounters are scored by the residue *res* — the unmatched geometry after complementary pairing — against the saturation threshold *TH* (§8.6):

- $res = 0 \rightarrow$ attraction. Full complementary cover; the encounter score. (Distinct from the later incorporation event, §4.6.)
- $0 < res \leq TH \rightarrow$ repulsion. Sub-threshold trapped geometry is rejected, not decayed.
- $res > TH \rightarrow$ forced decay. The saturated band selects forced decay per-tick; the multi-tick sequence is deferred to Paper 13.

FIGURE 4 — THE INTERACTION TAXONOMY



Attraction = scored verdict ($res = 0$); incorporation is the separate §4.6 alteration step.

6b THE RESIDUE MODEL (§8.6)

The residue *res* quantifies unmatched geometry after complementary pairing. Given an encounter's presented tuples, three independent computations agree (F.36 count formula, greedy pairing, maximum bipartite matching):

- $\text{res} = 0$: tick-tau full complementary cover — every presented value finds its complement. Clean resolution; the encounter scores attraction.
- $0 < \text{res} \leq \text{TH}$: trapped but sub-threshold — the geometry compounds but does not saturate. Scores repulsion; the junction is rejected, not decayed.
- $\text{res} > \text{TH}$: saturated — the trapped residue exceeds what the junction can hold. Scores forced decay (per-tick selection; multi-tick sequence deferred to Paper 13).

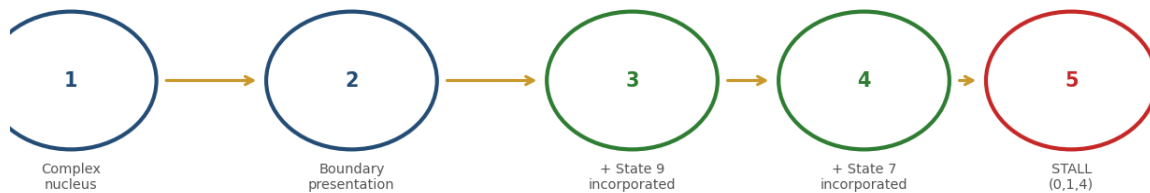
TH is the saturation threshold — the fuse, not a dial (Paper 7 §5.1.1). The saturated/middle boundary tracks TH across the viable band (F.53 TH-sensitivity).

7 CASE STUDIES

7.1 Case A — the nucleus→atom ladder (§9)

Nine simple states run first-encounter against nucleus (a): 27/27 attract ($\text{res} = 0$). The departure-ordered sequential chain incorporates States 9 and 7, then stalls — incorporation depletes the complementary inventory ((6,6,3) → (3,3,3) → (0,1,4)); subsequent encounters select forced decay. The 9! order-robustness sweep (1,088,640 chains) finds zero all-attraction orders; chains stall at 1–3 incorporations. The sequential all-attraction reading is falsified (§9.6). What is claimed, at Level B: first-encounter robustness.

FIGURE 5 — THE NUCLEUS→ATOM LADDER (WORKING CONSTRUCTION)



Case-A working construction, not a native sequential dynamics law. Sequential all-attraction falsified (§9.6).

7.2 Case B — collider outputs under forcing (§10)

Two regimes: clean resolution to simple states (the standard account's "elementary particle production"), and trapped complex overflow. The trapped band splits per §8.6: saturated ($\text{res} > \text{TH}$) selects forced decay; middle band ($0 < \text{res} \leq \text{TH}$) selects repulsion — rejected, not decayed.

7.3 Case C — complexity and manipulation paths (§11)

Lithium vs sodium battery chemistry: the configuration's complexity determines the set of consolidation paths available — six theorems (§11.2) on path upper bounds, monotonicity refutation, and boundary extension. Illustrative, not quantitative.

7.4 What the cases jointly establish

The three cases are independent cuts through the framework: Case A tests incorporation mechanics, Case B tests the forcing taxonomy, Case C tests manipulation-path structure. None depends on the others' conclusions. Jointly they exercise every branch of the §8.6 taxonomy and the full port-geometry machinery.

7c WORKED MINI-EXAMPLE

Encounter: State 7 (X,X,O) presents at an open port against nucleus (a)'s boundary. The scan presents the junction values at tick τ ; u is read off: two X-values unmatched $\rightarrow u=2 > 0$, so State 7's aspect is active (seeking). The nucleus presents complementary Y-values at $u=0 \rightarrow$ passive (reference). Complementary pairing: $X \leftrightarrow Y$ pairs zero out via T ($\Delta\tau_J = -2$ per tick) until $\tau_J = 0$ (aligned) or the residue stalls. If $\text{res}=0$: attraction scores; the junction may then incorporate (the separate §4.6 alteration step).

8 THE COMPLEXITY ORDERING

Complexity Ordering [LEVEL A] (§12). photon $\rightarrow$ open tilted states $\rightarrow$ meson-like $\rightarrow$ baryon-like $\rightarrow$ nuclei $\rightarrow$ atoms, ordered by axes flipped — a structural ordering of departure from baseline, not a particle claim and not a mass derivation. Machine-checked in exact integer arithmetic, conditional on the declared port-partition and aspect-attribution rules.

FIGURE 6 — THE COMPLEXITY ORDERING

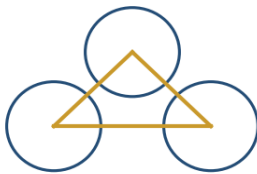


Structural ordering of axes flipped — not a particle hierarchy, not a mass derivation (§12.3).

The three worked aggregates — (a) symmetric three-node, (b) asymmetric two-node, (c) linear chain — verify the per-aspect s^2 decomposition under the declared Aspect Aggregation rule.

FIGURE 7 — THE THREE WORKED AGGREGATES

(a) symmetric three-node



(b) asymmetric two-node



(c) linear chain



Per-aspect decomposition under the declared Aspect Aggregation rule (not machine-derived).

9 LEVEL A CLAIM DIGEST (50)

Every claim below is proved (machine-checked, inherited, or prose-proved) with its lineage declared.

- STRUCTURAL_INSULATION_THEOREM (§4.1): The 27-state inventory is closed under the ledger.
- PORT_PARTITION_EXACTNESS (§4.3): Ports partition exactly; at most one completed junction per port. Native a_k derivation.

- PORT_COUNT_INVARIANCE (§4.3): Every node has exactly six ports; $B(G) = 6N - \sum k_j$.
- BOUNDARY_PORT_CHARACTERIZATION (§4.3): Boundary ports characterized by presented values.
- COMPOUND_CLOSED_LOOP_COHERENCE (§4.4): Per-aspect dissolution of the resolved/unresolved tension.
- RELATIONAL_ROLE_THEOREM (§3.1): No intrinsic active/passive; roles are (aspect, tick, encounter) functions. F.39.
- ROLE_PERMANENCE_PROHIBITION (§3.3): Roles cannot persist un-re-derived across ticks.
- CAPABILITY_ACTUALITY_SEPARATION (§3.1): Capability ("can play") vs actuality (encounter-indexed).
- SIMPLE_STATE_INSULATION (§5.2): Direct application of structural insulation.
- SIMPLE_STATE_REFERENCE_ROLE (§5.3): Saturated-center simple states (1, 9) can play the passive role. F.39.
- DUAL_RESPONSE_MODES (§6.3): Bonding modes 1 (outer-to-center) and 2 (outer-to-outer).
- SHOVE_TRANSITION_OPERATOR (§7.5): T zeroes the canonically-first pair; pairwise scope. F.43.
- TENSION_MEASURE_IDENTITY (§7.1): Shear tension IS the unmatched-integer count (definitional).
- TRACKING_COST_LEMMA (§7.3): Port-level s^2 contribution; one T-action yields non-positive Δs^2 . F.44.
- RELIEF_REDUCES_DEPARTURE (§7.3): Relief strictly reduces total departure.
- ASPECT_AGGREGATION (§7.3): Author-authorized: port contributions attributed to exactly one aspect.
- ALIGNMENT_DEFINITION (§7.2): Aligned iff tension zero; "toward" = per-tick direction.
- TRIPLE_CONVERGENCE_RESOLUTION (§7.6): Inherited (Paper 2 §8.1): $3 \times \text{State-4}$ yields photon/electron/positron.
- PROXIMITY_SCALING (§8.5): Engagement-weighted allocation; shell denominator carries d-dependence. F.45.
- PROXIMITY_SHELL_PROFILE (§8.5): Per-shell met-position profile; D-invariant 3/1/1. F.45.
- ENGAGEMENT_ALTERATION_SEPARATION (§4.6): Engagement vs alteration are distinct ledger operations. F.47.
- SCALE_FIDELITY (§0.5.v.3): Universality holds at all scales (N-threshold sweep). F.56.
- SINGLE_REGIME (§0.5): One substrate; no regime transitions natively. F.41.
- PHOTON_BASELINE (Paper 2 §5): Inherited: the photon is the universal ruler.
- INTERACTION_TAXONOMY (§8.6): The res/TH trichotomy. F.36.
- AGGREGATE_A/B/C_DECOMPOSITION (§4.1): Three worked aggregates; s^2 decomposition verified. F.33-35.
- CONSOLIDATION_PATHS (§11.2): Six theorems on manipulation path structure. F.48.

Level A registry/audit claims (14): R1-R23 rules (F.38), firewall groups (F.42), hygiene gates (F.49). Level B (4): case-study illustrations. Level C (3): Paper-12/13/16 deferrals.

10 KEY PROOF SKETCHES

Port-Partition Exactness (§4.3)

Three cases: (1) distinct a_k entries — "no longer open" persistence (§6.1); (2) same entry — Port Status binary; (3) intra-junction duplicates — excluded by definiens. Two scans cannot allocate the same finite contribution twice.

Relational Roles (§3.1)

F.39 exhibits one selector active ($u>0$) in R1 and passive ($u=0$) in R2 — identical selector, only port values change. Intrinsicity refuted.

Proximity Scaling (§8.5)

Shell denominator $N(d)=4d^2+2$ inherited (Paper 10 §5.3.1); engagement count derived per-shell from presented geometry (D-invariant 3/1/1). The v46 defect repaired.

11 FALSIFICATION CONDITIONS

- C19. A role observed persisting across ticks without re-derivation voids Role Permanence.
- C20. A port hosting two completed junctions voids Port-Partition Exactness.
- C21. Entity-intrinsic role labels (not encounter-indexed) trigger falsification.
- C22. Double-counted port or unbalanced ledger voids the machine-check half.

11b DEPENDENCY LABELS (THE §0.5 ARCHITECTURE)

Every theorem carries a bracketed badge listing the architectural layers its proof leans on:

- L1 — The 27-state inventory (3-axis count).
- L2 — Hardware topology axiom (6-port array).
- L3 — Relay-step metric axiom.
- L4 — Null boundary default (no correction terms).
- L5-P11 — Relational role theorem (derived in Paper 11).
- L6-P11 — Structural-cut definitions (simple vs complex).
- L7-P11 — Operational role-cut definitions (active/passive).
- L8-L10-P11 — Encounter geometry, budget, and tick definitions.

Bridge rule (Paper 10.5): a downstream claim may not carry a stronger epistemic level than its weakest necessary bridge — $L_{\text{claim}} \leq \min(L_i)$.

13 BRIDGE LEDGER (IMPORTS / EXPORTS)

Imports (inherited, not re-proved): Paper 2 (27 states, a_k , bonding modes, triple-convergence); Paper 3 (inside-outside ledger, 9-state stability); Paper 5 (closure principle); Paper 6 (ledger states); Paper 7 (forcing/fuse); Paper 8 (mass accounting, circa budget); Paper 10 (gravity, shell count $N(d)=4d^2+2$).

Exports (for downstream papers): the role vocabulary (§13.2, sequential-construction ban — constrains Papers 14–16); the complexity ordering (Paper 12 bulk regimes); the multi-tick deferral (Paper 13 time formalism); meson/baryon mapping (Paper 16).

Firewalls: Paper 12 owns many-body/stat-mech/bulk thermo — Paper 11 stays at configuration-role level. Paper 13 owns the temporal formalism — Paper 11 verifies per-tick selections only.

14 KEY REFERENCES

- Paper 2 §1 — a_k tick primitive; §4.1 — three-axis fabric; §5 — photon baseline; §8.1 — triple-convergence; §11.1 — bonding modes / saturated centers; §12 — ledger cumulativity.
- Paper 3 §2 — inside-outside ledger; §7–§8.5 — convergence geometry and 9/9/9 balance.
- Paper 5 §2 — closure principle; §4 — nine-state stability ranking.

- Paper 6 — rest/uniform motion/inertia as ledger states.
- Paper 7 §5.1.1 — threshold-as-fuse; forcing.
- Paper 8 — mass as accounting of departure; circa budget.
- Paper 10 §2 — gravity as resolution-seeking; §2.2 — circa budget; §5.3.1 — shell count $N(d)=4d^2+2$; §7.1 — forcing categories.
- Paper 10.5 — bridge ledger methodology ($L_{\text{claim}} \leq \min(L_i)$).

14b APPENDICES (MAP)

What lives where in the full paper:

- Appendix E — the machine-readable claim registry (E.2): all 57 claims with IDs, levels, lineage, and verification pointers. The auditor's source of truth.
- Appendix G — the figure plan: all 7 figures with captions and what each depicts.
- Appendix H — audit rules R1-R23: the self-test specification (claim-impact statements, axiom rules, closure rules, register integrity).
- Appendix I — revision history: the complete changelog (#1-#147), every remediation and its evidence.
- Appendix J — theorem map: dependency graph of all Level-A theorems (F.52 generated).

16 COMPLETENESS

Staged vs asserted: F.54 (proof-registry crosscheck) and F.55 (alignment correspondence) are staged — listed, not wired. Everything else is asserted and green.

Version v69 (2026-10-07) · 28/28 scripts exit 0 · 57 claims (A50/B4/C3) · All listings byte-identical · 7 figures rendered · Fences even.

Falsification conditions C19-C22 (§14.3): the paper states exactly what observations would void its core claims — entity-intrinsic role labels (C21), port double-allocation (C22), and the compound-closed-loop machine-check halves.